

Digital Forensics Project Part 1

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Executive Section

Our investigation reveals a concerning breach in the security of our network, resulting in unauthorized access and data exfiltration. The incident occurred on September 18, 2020, at 10:24 PM EDT, when an attacker entered our server (DC01) and subsequently accessed sensitive files, including the CA root certificate and the Szechuan sauce recipe. The attacker utilized Remote Desktop Protocol (RDP) as the initial entry vector and deployed malware (coreupdater.exe) to facilitate their malicious activities. Additionally, the attacker stole the favicon.ico file from the desktop on September 19, 2020, at 02:23:41 AM EDT. Notably, the last known contact with the adversary involved the exchange of encrypted application data from our server to the attacker's IP address (194.61.24.102).

Technical Findings

When was the system compromised?

- Based on the case001 pcap file, the DC01 - E01 image file, the server or system seemed to have been compromised on September 18, 2020, at 10:24 PM, starting with an unauthorized Remote Desktop Protocol connection.

How was it compromised?

From the evidence collected, the system was compromised through a series of steps involving initial entry vector, Utilization of Malicious Software, Persistence and Lateral Movement, Data Compromise, Network Setup and Security Enhancements and Final Communication with the Adversary. Here's a more detailed explanation of how the system was compromised through Remote Desktop Protocol (RDP):

1. Initial entry vector:

- The unauthorized access to the system occurred via Remote Desktop Protocol (RDP), a Microsoft-developed proprietary protocol enabling remote connections between computers over a network. It's likely that the attacker exploited vulnerabilities or weak login credentials to remotely access the server (DC01).

2. Utilization of Malicious Software:

- Following the breach, the attacker introduced coreupdater.exe, a malware tool, onto the server. This malicious software was instrumental in carrying out various harmful actions on both the server and the desktop.
- The delivery of coreupdater.exe originated from the IP address 194.61.24.102, pinpointing the source of the malicious payload.

3. Persistence and Network Exploration:

- The coreupdater.exe malware was implanted with persistence on both the server and the desktop, ensuring its automatic activation during system startup. This persistence mechanism granted the attacker continual access to the compromised systems even after system reboots.
- Presence of the malware on both server and desktop hints at lateral movement within the network. Once inside, the attacker navigated from the server to the desktop, potentially seeking to elevate privileges or obtain further resources.

4. Data Compromise:

- The breach involved the theft of sensitive files from the server, including critical documents like the CA root certificate and the Szechuan sauce recipe. This suggests that the attacker's motive encompassed data theft, potentially for financial gain or other nefarious purposes.

5. Network Setup and Security Enhancements:

- The victim network comprises a Windows Server 2012-based server (DC01) and a Windows 10 desktop (Desktop-sdn1rpt). Both machines are interconnected within a local network, presumably safeguarded by firewall and router protection.
- Immediate measures should prioritize revoking the compromised CA root certificate and issuing a new one to thwart any further unauthorized usage. This proactive step aims to minimize risks associated with the compromised certificate.

6. Final Communication with the Adversary:

- The most recent interaction with the adversary involved the transmission of encrypted application data from the server to the attacker's identified IP address (194.61.24.102). This ongoing communication underscores the urgency for prompt remedial action and the necessity for deeper investigative efforts.

Who compromised the system (attacking IP address)?

Based on the evidence we've gathered; it seems that the system with the IP address 10.42.85.10 was compromised through a connection using Remote Desktop Protocol (RDP). The attacking IP address, 194.61.24.102, was found to be the source of this unauthorized access and malicious activity within our network.

The attacker managed to break into our network by exploiting the Remote Desktop Protocol (RDP). They then introduced a harmful software called coreupdater.exe onto both our server (DC01) and our desktop (Desktop-SDN1RPT). This malware, originating from the IP address 194.61.24.102, had the ability to access our systems remotely, steal

data, and manipulate system settings. It was designed to stay hidden and keep working even if our systems were restarted.

Once inside our network, the attacker moved on to access other systems, likely taking advantage of the compromised RDP access they already had. They first got into our DC01 using an Administrator Account and then moved on to the desktop.

The attacker didn't stop there. They went on to steal sensitive information from both our DC and desktop, including the CA ROOT certificate and zip files. This suggests that they were after valuable data for malicious purposes.

The IP address 194.61.24.102 was identified as belonging to a known group of cyber attackers. Additionally, we found another suspicious IP address, 205.185.216.10, which was involved in similar attacks around the same time.

What did they do?

1. Unauthorized Access to Sensitive Files: The attacker managed to breach security measures and gain illicit entry to sensitive files stored on both the server (DC01) and the desktop (Desktop-SDN1RPT). Among these files were critical documents such as the CA root certificate and the recipe for Szechuan sauce. These files likely contained confidential or valuable information vital to the organization's operations.

2. Introduction of Malicious Software: The attacker planted malware known as coreupdater.exe within the victim network. This malicious program was surreptitiously installed on both the server and the desktop, providing the attacker with a means to execute a variety of nefarious actions from a remote location. The malware equipped the attacker with capabilities for remote access, data theft, and manipulation of system settings.

Did they access folders/files?

- Yes, the attacker accessed sensitive files, including the CA root certificate and the Szechuan sauce recipe, signifying a targeted effort to obtain valuable information.

Did they use specific tools?

- Yes, the attacker employed malware (coreupdater.exe) as a tool to facilitate their malicious activities, allowing for remote access and data manipulation

Did they run programs/scripts?

- Yes, the attacker executed malicious programs, likely in the form of scripts associated with the deployed malware, to gain remote access and steal data from the compromised systems.

Conclusion

In conclusion, the forensics investigation into the cyber attack on the Szechuan sauce. The investigation into the security breach reveals a troubling compromise within our network, leading to unauthorized access and the exfiltration of sensitive data. The incident, occurring on September 18, 2020, at 10:24 PM EDT, initiated through a Remote Desktop Protocol (RDP) connection, facilitated the infiltration of our server (DC01) and subsequent access to critical files, including the CA root certificate and the Szechuan sauce recipe. The attacker's actions encompassed several stages, including the exploitation of RDP for initial access, the deployment of coreupdater.exe malware for remote manipulation, and the extraction of sensitive information from both the server and desktop. Furthermore, the persistence of malware on compromised systems, lateral movement within the network, and ongoing communication with the adversary underscore the severity and sophistication of the attack.